

LEIPZIG HOLZHAUSEN, Germany

WMO: 104710

Lat: 51.3150N

Lon: 12.4461E

Elev: 453

StdP: 14.46

Time Zone: 1.00 (EUC)

Period: 97-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.2	17.2	7.2	8.1	14.3	11.9	10.3	19.7	18.3	48.0	16.3	44.9	4.7	80	0.477

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.4	86.5	66.8	83.0	65.2	79.6	64.2	68.8	81.2	67.3	78.3	65.8	76.0	5.2	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.9	93.8	72.0	63.4	89.0	70.9	62.0	84.6	69.9	33.2	81.2	31.9	78.5	30.8	76.2	75.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.5	12.7	11.3	7.5	93.7	6.7	3.5	2.6	96.2	-1.3	98.2	-5.0	100.2	-9.9	102.8
			6.8	71.9	6.6	1.5	2.0	73.0	-1.8	73.9	-5.5	74.8	-10.3	75.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	50.1	34.1	35.2	41.1	50.1	58.0	63.0	66.6	66.4	59.0	50.1	41.8	35.0	(6)
(7)		DBStd	13.87	9.34	8.97	7.67	7.21	6.62	6.30	5.71	5.55	5.94	6.83	6.59	7.75	(7)
(8)		HDD50	2103	494	417	287	86	10	1	0	0	3	85	255	465	(8)
(9)		HDD65	5704	957	833	741	448	235	111	47	49	195	462	697	929	(9)
(10)		CDD50	2148	1	4	12	89	257	390	516	508	274	88	8	1	(10)
(11)		CDD65	276	0	0	0	2	17	50	98	92	16	1	0	0	(11)
(12)		CDH74	2388	0	0	0	21	181	447	886	716	135	2	0	0	(12)
(13)		CDH80	675	0	0	0	3	35	117	298	205	17	0	0	0	(13)
(14)	Wind	WSAvg	5.2	6.1	6.1	6.0	5.2	5.1	4.6	4.6	4.5	4.4	5.0	5.1	5.9	(14)
(15)	Precipitation	PrecAvg	20.3	1.3	1.1	1.4	1.5	2.0	2.2	2.4	1.8	1.4	1.5	1.5	(15)	
(16)		PrecMax	29.8	3.1	3.2	3.8	4.5	4.9	4.4	6.1	6.2	5.5	5.2	4.0	3.3	(16)
(17)		PrecMin	11.9	0.0	0.1	0.2	0.1	0.2	0.4	0.3	0.5	0.2	0.1	0.0	0.0	(17)
(18)		PrecStd	4.3	0.7	0.6	0.8	0.9	1.1	0.9	1.3	1.3	1.0	1.0	0.8	0.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.5	58.0	65.5	76.3	84.3	88.5	92.7	91.1	82.2	71.8	60.1	55.0	(19)
(20)		MCWB	49.0	49.8	52.7	59.2	63.9	68.8	68.6	66.7	64.5	61.5	53.5	49.3	(20)	
(21)		2%	DB	52.0	53.7	60.7	71.5	79.3	83.5	87.6	85.1	77.9	66.4	56.1	50.7	(21)
(22)		MCWB	47.1	48.1	50.9	56.7	62.7	66.0	66.4	66.5	62.7	58.2	50.8	46.2	(22)	
(23)		5%	DB	49.2	50.5	56.2	67.4	74.6	79.1	82.9	81.4	73.9	63.1	53.1	47.9	(23)
(24)		MCWB	45.1	45.5	48.3	54.7	60.3	64.2	65.5	64.7	61.4	56.4	48.9	44.3	(24)	
(25)		10%	DB	46.1	47.6	52.6	63.4	70.6	75.4	78.7	77.7	69.9	60.1	50.9	45.4	(25)
(26)		MCWB	42.6	43.6	46.5	52.4	58.0	62.6	64.4	63.9	60.1	54.8	47.3	42.4	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.4	50.4	54.0	61.1	66.6	71.8	70.4	70.3	66.5	62.0	54.3	50.1	(27)
(28)		MCDWB	54.4	56.0	62.3	75.1	80.3	82.3	85.4	84.8	77.5	68.9	59.7	53.7	(28)	
(29)		2%	WB	47.8	48.7	51.3	57.9	64.1	68.0	68.6	68.6	64.3	59.3	51.3	46.6	(29)
(30)		MCDWB	51.2	53.2	58.3	68.5	75.4	79.8	80.8	80.6	73.7	65.1	54.9	49.8	(30)	
(31)		5%	WB	45.4	46.2	49.4	55.8	62.0	65.9	67.1	66.9	62.6	57.1	49.6	44.6	(31)
(32)		MCDWB	48.8	49.8	55.6	65.3	71.8	76.2	78.2	76.8	70.8	61.8	52.7	47.8	(32)	
(33)		10%	WB	42.8	43.6	46.9	53.5	59.8	63.8	65.6	65.2	60.9	55.2	47.7	42.4	(33)
(34)		MCDWB	45.8	47.6	52.0	61.9	68.6	72.7	76.2	74.2	68.8	59.5	50.6	45.2	(34)	
(35)	Mean Daily Temperature Range	MDBR	8.7	10.0	13.0	17.1	17.8	17.1	17.4	17.4	16.0	13.0	9.1	7.8	(35)	
(36)		5% DB	MCDBR	10.5	12.9	19.9	24.7	24.5	24.0	25.5	24.0	23.2	17.3	11.9	9.9	(36)
(37)		MCWBR	7.9	8.9	11.8	12.3	10.9	9.7	8.7	8.8	10.4	10.3	8.2	7.6	(37)	
(38)		5% WB	MCDBR	10.0	11.0	17.5	22.2	21.6	21.7	21.5	19.9	19.4	16.0	11.0	9.9	(38)
(39)		MCWBR	8.0	8.1	11.0	11.6	10.4	9.7	8.3	8.5	9.3	10.1	8.1	7.8	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.316	0.340	0.384	0.418	0.420	0.416	0.434	0.427	0.389	0.366	0.339	0.310	(40)	
(41)		taud	2.364	2.330	2.227	2.181	2.212	2.252	2.210	2.248	2.334	2.373	2.388	2.366	(41)	
(42)		Ebn at Noon	221	246	254	259	265	266	259	254	251	233	209	202	(42)	
(43)		Edh at Noon	20	27	36	42	43	41	43	39	32	26	20	17	(43)	
(44)	All-Sky Solar Radiation	RadAvg	236	459	776	1266	1544	1654	1564	1355	975	567	271	181	(44)	
(45)		RadStd	33	76	101	162	185	137	193	121	125	67	38	20	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (52 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.59	N/A	N/A	N/A	+40

Nomenclature: See separate page